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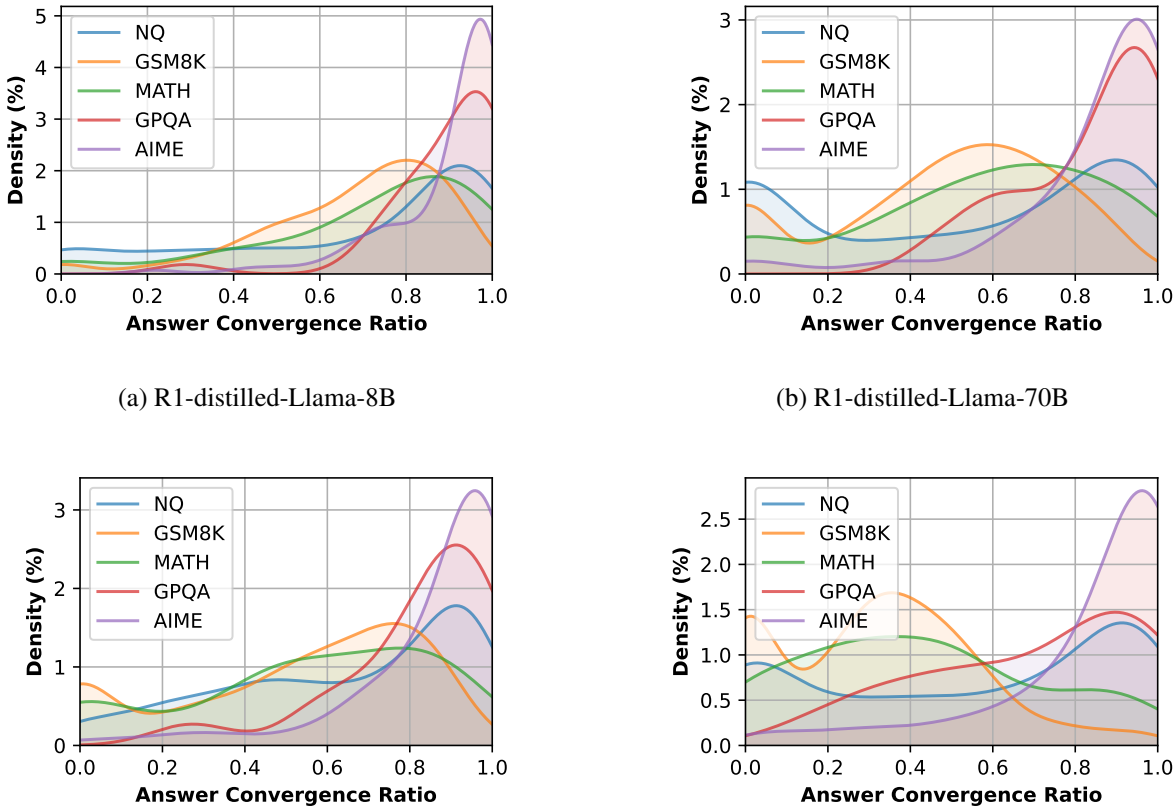
A Appendix

A.1 ACR Distribution on Various Models

We present the ACR distributions for several models: R1-distilled Llama-8B, R1-distilled Llama-70B, R1-distilled Qwen-7B, and QwQ-32B. The results are shown in Figure 4. All models exhibit similar trends to the R1-distilled Qwen-32B baseline, confirming their ability to produce stable answers before completing the full reasoning chain. Moreover, tasks with higher reasoning demands tend to correspond to higher ACRs. When comparing distributions across models, we observe that larger models generally achieve lower ACRs, suggesting that they require fewer reasoning steps to converge on an answer. This implies that larger models may possess a more efficient internal reasoning process, enabling them to reach final answers more quickly.

A.2 Dataset Splits and Statistics

For tasks with available training data (NQ and GSM8K), we sample 1,000 examples for training the supervised method and 100 for validation. For



(a) R1-distilled-Llama-8B

(b) R1-distilled-Llama-70B

(c) R1-distilled-Qwen-7B

(d) QwQ-32B

Figure 4: ACR distributions of various models.

Tasks	Train	Validation	Test
NQ	800	200	3610
GSM8K	800	200	1319
MATH-500	320	80	100
GPQA-Diamond	78	20	100
AIME'24	–	–	30

Table 2: Dataset statistics of each task.

tasks without predefined training data (MATH-500 and GPQA-Diamond), we reserve 100 examples as the test set and split the remaining data into 80% for training and 20% for validation. Due to the limited size of AIME'24, which includes only 30 test examples, we evaluate only the unsupervised methods on this task. The statistics for each task are shown in Table 2.

A.3 Implementation of Early Stopping via Boosting End of Think Token

To implement boosting of the `</think>` token, we design a logit processor that adjusts the model's output logits before sampling. Specifically, it increases the logit of the `</think>` token according to §4.2. Once the `</think>` is generated, the processor is disabled to prevent further modifications. This mechanism allows the model to emit the `</think>` token earlier if it has already reached a confident answer. We integrate the logit processor into the VLLM framework (Kwon et al., 2023), enabling efficient logit manipulation during decoding without compromising throughput. We run all experiments three times and report the average results.

A.4 Implementation Details

When applying early stopping via answer consistency, we empirically set the number of consecutive chunks k to 10. For early stopping via boosting, we set the hyperparameter α to 0.6. Regarding the supervised method, we use a single-layer LSTM with 128 hidden units and a dropout rate of 0.1. The model is trained for 200 epochs with a batch size of 32, using the Adam optimizer with a learning rate of $5e^{-4}$. The confidence threshold τ is tuned on the validation set, and we set it to 0.50, 0.99, 0.99, and 0.50 for NQ, GSM8K, MATH-500, and GPQA-Diamond, respectively. For the CCoT baseline, we set the token budget to 100 for all tasks. We run all experiments three times and report the averaged results.

We use the VLLM framework (Kwon et al., 2023) to sample the model outputs for all experiments to ensure efficient inference. All the prompts we used are provided in Appendix A.8.

A.5 Ablation Study

Setup. We ablate two key hyperparameters used by our unsupervised methods: (i) α for *Think Token Adjustment*, which up-weights the probability of the end-of-reasoning marker; and (ii) k for *Answer Consistency*, the consecutive-consistency threshold. Unless otherwise noted, we report results with R1-distilled-Llama3.1-8B across five benchmarks (NQ, GSM8K, MATH-500, GPQA, AIME'24).

A.5.1 Effect of α (Think Token Adjustment)

Accuracy. Moderate boosting ($\alpha \leq 0.4$) maintains accuracy on easier tasks while larger α can cause premature termination and degrade performance on harder reasoning tasks.

α	NQ	GSM8K	MATH-500	GPQA	AIME'24
0.0	20.6	78.6	79.0	14.1	40.0
0.2	23.2	74.6	79.0	14.1	40.0
0.4	24.6	73.2	73.0	14.1	40.0
0.6	25.2	66.8	71.0	9.1	23.3
0.8	20.4	31.2	50.0	8.1	30.0
1.0	15.8	43.2	67.0	8.1	30.0

Table 3: Effect of α on **accuracy** (%).

Token Usage. As α increases, tokens drop sharply; however, overly large α harms accuracy despite aggressive savings.

α	NQ	GSM8K	MATH-500	GPQA	AIME'24
0.0	426.8	401.3	2011.8	2546.0	7914.4
0.2	426.8	366.4	2011.8	2546.0	7914.4
0.4	418.9	349.2	1712.5	2393.9	5060.0
0.6	418.5	316.1	1017.9	1557.0	2429.8
0.8	91.3	49.1	68.5	79.1	81.4
1.0	17.6	10.3	11.2	27.8	9.3

Table 4: Effect of α on **#tokens**.

Takeaway. $\alpha \in [0.2, 0.4]$ offers a good accuracy–efficiency trade-off; aggressive boosting ($\alpha \geq 0.6$) risks early truncation on hard tasks.

A.5.2 Effect of k (Answer Consistency)

Accuracy. Small k stops too early and hurts accuracy; performance improves and saturates around $k \in [20, 30]$.